

COMPUTER EXERCISE 2

VALENCIA 2019

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1. EXERCISE 1

Given the data, `y <- c(4, 5, 4, 5, 3, 1, 2, 3, 4, 8, 0, 1, 4, 0, 2)`, we need to figure out from which Poisson distribution the data has been generated i.e. for which λ of $Po(x; \lambda)$

A good guess is the sample average.

```
> y <- c(4, 5, 4, 5, 3, 1, 2, 3, 4, 8, 0, 1, 4, 0, 2)
> y_bar <- mean(y)
> y_bar
[1] 3.066667
```

A more theoretical solution is to use the Poisson density function,

$$Po(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Given $X_i \sim Po(x; \lambda), i = 1, 2, \dots, n$, the log-likelihood function,

$$\begin{aligned} l(\lambda; x) &= \sum_{i=1}^n \left(x_i \log(\lambda) - \lambda - \log(x_i!) \right) \\ &= \left(n \log(\lambda) \right) \bar{x} - n\lambda - \sum_i \log(x_i!). \end{aligned}$$

Then

$$\frac{dl}{d\lambda} = 0 \Leftrightarrow \bar{x} = \lambda.$$

. Thus given the data, the theoretical mean is 3.066666666666667.

2. EXERCISE 2

We now use the `glm` function to estimate λ .

```
> y_data <- data.frame(y)
> glm1 <- glm(y ~ 1, family = poisson(link = "log"), data = y_data)
> summary(glm1)
```

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Call:

```
glm(formula = y ~ 1, family = poisson(link = "log"), data = y_data)
```

Deviance Residuals:

	Min	1Q	Median	3Q	Max
	-2.47656	-1.01318	-0.03821	0.50888	2.33986

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.1206	0.1474	7.6	2.96e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 25.455 on 14 degrees of freedom
 Residual deviance: 25.455 on 14 degrees of freedom
 AIC: 66.629

Number of Fisher Scoring iterations: 5

```
> beta0 <- summary(glmmod1)$coefficients[, 1]
> estimate<-exp(beta0) # estimate of lambda
```

The estimate of 3.06666666666668 agrees with the theoretical estimate.

Get the model diagnostic plots using the following code:

```
> plot(glmmod1)
```

3. EXERCISE 3

We fit the model:

```
> y <- c(0,0,0,0,0,0,0,0,0,0,0,1,1,1,1,1,1,1,1,1)
> x <- c(0,1,1,0,0,1,0,1,0,1,2,2,0,1,1,2,1,1,2,1)
> glm.CC <- glm(y ~ x, family=binomial)
> summary(glm.CC)
```

Call:

```
glm(formula = y ~ x, family = binomial)
```

Deviance Residuals:

	Min	1Q	Median	3Q	Max
	-1.28168	-0.69376	-0.05118	1.07659	2.07345

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.026	1.111	-1.823	0.0684 .
x	2.268	1.077	2.105	0.0353 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 27.726 on 19 degrees of freedom
 Residual deviance: 20.172 on 18 degrees of freedom
 AIC: 24.172

Number of Fisher Scoring iterations: 4

The p value for the SNP effect is 0.0353 and we may conclude that the effect is significant.

4. EXERCISE 4

We fit the model using the code given in the exercise

```
> library(faraway)
> data(savings) #See Faraway page 29
> g <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data=savings)
> summary(g)

Call:
lm(formula = sr ~ pop15 + pop75 + dpi + ddpi, data = savings)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-8.2422	-2.6857	-0.2488	2.4280	9.7509

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	28.5660865	7.3545161	3.884	0.000334 ***
pop15	-0.4611931	0.1446422	-3.189	0.002603 **
pop75	-1.6914977	1.0835989	-1.561	0.125530
dpi	-0.0003369	0.0009311	-0.362	0.719173
ddpi	0.4096949	0.1961971	2.088	0.042471 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.803 on 45 degrees of freedom

```

Multiple R-squared:  0.3385,          Adjusted R-squared:  0.2797
F-statistic: 5.756 on 4 and 45 DF,  p-value: 0.0007904

> res1 <- residuals(g)
> hv1 <- hatvalues(g)
> response1 <- (res1^2)/(1-hv1)
> weight1 <- (1-hv1)/2
> glm1 <- glm(response1 ~ 1, weights = weight1, family=Gamma(link=log))
> summary(glm1)

Call:
glm(formula = response1 ~ 1, family = Gamma(link = log), weights = weight1)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.13709  -1.25655  -0.42999   0.00425   1.95326

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.6714     0.2004   13.33  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.9037061)

Null deviance: 51.555  on 49  degrees of freedom
Residual deviance: 51.555  on 49  degrees of freedom
AIC: 162.94

```

Number of Fisher Scoring iterations: 7

The fitted linear model gives a residual standard error of 3.8027 the residual variance is the squared value, i.e. 14.4603. The glm output gives the estimated residual variance on the log scale 2.6714 and its standard error is 0.2004. A 95 % CI is computed as $\exp(2.6714 - 1.96 \cdot 0.2004)$ and $\exp(2.6714 + 1.96 \cdot 0.2004)$, which gives the interval [9.763, 21.4176].

5. EXERCISE 5

The function below fits a Poisson null model with canonical link using an Iterative Reweighted Least Squares algorithm. The data from exercise 1 is fitted with the given function and compared with `glm`. As can be seen, the results are identical.

```

> #---- IWLS for Poisson null model with log-link ----#
> PoissonGLM <- function(y, inita = 0.01, maxit = 100) {
+   b <- c(inita)
+   X <- cbind(rep(1,length(y)))
+   for (i in 1:maxit) {
+     eta <- X%*%b
+     mu <- exp(eta) # iverse of the link function
+     deta_dmu <- 1/mu # d eta/d mu
+     v_mu <- mu # V(mu)
+     z <- eta + (y - mu) * deta_dmu # Working response
+     w <- (deta_dmu)^2 * (v_mu) # Weights
+     lmod2 <- lm(z ~ 1, weights = 1/w) # Run weighted regression
+     cf <- coef(lmod2)
+     if (max(abs(b - cf)) < 1e-8) break
+     b <- as.numeric(cf)
+   }
+   if (i < maxit) {
+     return(cf)
+   } else {
+     return("Algorithm failed to converge")
+   }
+ }
> #----
>
>
> #---- Fit GLM model ----#
> glm3 <- glm(y ~ 1, family = poisson(link="log"))
> my_glm3 <- PoissonGLM(y = y)
> #---- Compare results ----#
> my_glm3
(Intercept)
-0.6931472
> coef(glm3)
(Intercept)
-0.6931472

```

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